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Original paper

Identification of the honey bee swarming process by analysing the time course of hive vibrations

Martin Bencsik, Joseph Bencsik, Michael Baxter, Andrei Lucian, Julien Romieu, Mathias Millet Show more

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Abstract

Honey bees live in groups of approximately 40,000 individuals and go through their reproductive cycle by the swarming process, during which the old queen leaves the nest with numerous workers and drones to form a new colony. In the spring time, many clues can be seen in the hive, which sometimes demonstrate the proximity to swarming, such as the presence of more or less mature queen cells. In spite of this the actual date and time of swarming cannot be predicted accurately, as we still need to better understand this important physiological event. Here we show that, by means of a simple transducer secured to the outside wall of a hive, a set of statistically independent instantaneous vibration signals of honey bees can be identified and monitored in time using a fully automated and non-invasive method. The amplitudes of the independent signals form a multi-dimensional time-varying vector which was logged continuously for eight months. We found that combined with specifically tailored weighting factors, this vector provides a signature highly specific to the swarming process and its build up in time, thereby shedding new light on it and allowing its prediction several days in advance. The output of our monitoring method could be used to provide other signatures highly specific to other physiological processes in honey bees, and applied to better understand health issues recently encountered by pollinators.

Research highlights

► An accelerometer in the wall of a honey bee hive supplies data to monitor activity. ► Unwanted external sounds are not picked up. ► PCA based feature extraction can identify a swarming-specific signal. ► This signal can predict the swarming event several days in advance. ► By storing the averaged spectra, logging of relevant long-term signal is feasible.

Keywords

Honey bee; Swarming; Vibrations; PCA

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Abstract

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Introduction

Honey bees live in societies of several thousands of individuals and form the object of much scientific work due to the complexity and sophistication of their behaviour. Following the major advances in this field brought by Von Frisch (1967), extensive work has been done focusing on further understanding the way honey bees communicate, in particular with acoustic noise and body vibrations.

In the waggle dance 280 Hz sounds are produced modulated by a 15 Hz body wag (Michelsen, 1999, pp. 111–131), performed directly on the comb (although not involving abdominal contact). It has been shown (Tautz, 1996) that the substrate on which the dance is performed affects the quality of the communication process (although plastic comb foundations do not (Seeley et al., 2005)), that bee legs are most sensitive to propagating (in the comb) longitudinal vibrations between 30 and 100 Hz (Sandeman et al., 1996), that vibrations found in the comb between 200 and 300 Hz are significantly enhanced during the waggle phase (Nieh and Tautz, 2000), and that phase reversal of the propagating wave occurs in close proximity to the dancer, at a distance at which followers are seen to be most frequently attracted to the messenger (Tautz et al., 2001).

Studies of sound radiated by bees have focused on the sound field of single bees (Michelsen et al., 1987), and it appears that follower bees tend to place their antennae in the zone of maximum acoustical short-circuiting. The properties of 'piping', a specific sound pulse which has been shown to be intimately linked with the preparation for swarming (Seeley and Tautz, 2001, Visscher and Seeley, 2007), has also been studied in terms of acoustic noise, both for 'worker piping' (Seeley and Tautz, 2001, Visscher and Seeley, 2007) and 'queen piping' (Kirchner, 1997, pp. 273–300). Measurements of the sound radiated by a collection of bees have been reported in the scientific literature (Ferrari et al., 2008), by placing a microphone on the top of the frames of a hive. This only provides crude and unreliable short term recordings, although the authors have some evidence that these can indicate the preparation for the swarming process. Similar microphone measurements are also found in a few patents (Bromenshenk et al., 2007, Bromenshenk et al., 2009, Etter et al., 2007) including exemplification data by which the inventors demonstrate that upon exposure to predatory mites or to sub-lethal concentrations of a few specific airborne toxicants, differing signatures may be extracted from the measured honey bee sound. A separate patent (Woods, 1957) claims that a set of simple band pass filters applied to the acoustic noise of bees may be a good indicator of the vicinity of the swarming process but no exemplification data is made available to the reader to substantiate the claim.

Very recently, a device for the long term logging of important features of honey bee sounds in a hive has also been successfully developed (Atauri and Llorente, 2009).

Those honey bees that are managed by humans often live in wooden hives, and these emit acoustic noise that can clearly be heard if the ear is located near enough to it. In periods of very high activity, the mechanical vibrations sustained in the hive's outer box can also be sensed simply by placing the hand on the wood.

In our work these vibrations were sensed by two accelerometers secured onto the outer wall of two separate hives, approximately 3 m away from one other, each comprising of a healthy colony of *Apis mellifera* honey bees. We are focusing on analysing this 'by-product' of honey bee activity, to show that it comprises of

complex global information, rather than taking the microscopic approach of exploring the insect's individual communication process.

We present one method for processing this raw data in order to extract a specific time varying signature comprising 10 independent components, and demonstrate that this can be applied to identify and predict the swarming process several days in advance.

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Honey bees under investigation

Two *Langstroth* hives were monitored in Jarnioux, France (latitude (DMS) 45°58′0N longitude (DMS) 4°37′60E altitude (meters) 271). They were located in close (10m) proximity to a house comprising a computer indoors (Pentium III 800MHz CPU running a Linux operating system). A simple roof was placed above the monitored hives to prevent vibrations caused by rain drops directly falling onto them. The two colonies were looked after with minimal interference, although they were fed during the winter. …

Spectrograms

The one day-long spectrograms of the two hives are shown in Fig. 1 for a specific day. For both hives four to five ‘bands’ are clearly seen, the strongest being around 2000Hz. Small frequency differences can be seen between the two hives, resulting from a combination of different hives and, perhaps, from differences coming from the honey bee colonies themselves. Continuous activity in the night can be seen as well as amplitude variations over two orders of magnitudes. The second hive provides …

Discussion and further work

We have shown that a non-invasive and automated method can provide to be a useful monitoring device, by analysis of the time course of the normal vibrations recorded on the wall of a honey bee hive. We chose to use accelerometers in our work, which, to our knowledge, have never been tried as hive monitoring sensors. Although not published in the professional literature, we know from our own experience and that of others (Michelsen, 2005) that microphone measurements are only sensitive to a few …

Acknowledgements

We are grateful to Dr. Robert Morris, NTU, who wrote the Linux script continuously storing the digitised data in individual short files, to Dr. Daniel Brown, NTU, who worked out the sun rise times from the location of the hives, to Antoine and Joseph Bencsik Jr. for looking after the 8-month long continuous recording, and to NTU who partly funded this work through a SPUR summer work bursary which Andrei Lucian benefited from. …

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Research article

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Edwards-Murphy F., ..., Popovici E.M.

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> ...This paper is organised as follows: Section 2 describes recent work in the area; Section 3 gives a system overview; Section 4 describes the first deployments; Section 5 presents the data analysis; Section 6 outlines the developed signal processing algorithms; Section 7 provides results and analysis of the developed decision tree performance; and Section 8 concludes the paper. Using technology to monitor honey bees has been the subject of extensive previous work, examples include detecting a hive state known as “swarming” by Bencsik et al. (2011) and Ferrari et al. (2008) and counting the number of bees exiting and entering a hive using LEDs by Chen et al. (2012). These systems all use fixed sensors connected to PCs, making them unsuitable for remote, large scale, or widespread deployments....

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